

YUHAN WU

yuhangyw@gmail.com | Singapore
[https:// www.linkedin.com/in/yuhang-wu-146749320](https://www.linkedin.com/in/yuhang-wu-146749320)

PERSONAL SUMMARY

Background in **Data Science, AI Agent Systems, and AI/ML applications**, with experience in anomaly detection, user behavior modeling, intelligent reporting, and computer vision / multimodal applications. Skilled at **applying advanced AI/ML techniques to complex real-world problems**, integrating data modeling, LLM-enabled workflows, and computer vision methods into practical solutions that support analysis, automation, and decision-making. Projects span game operations analytics, conversion modeling, robotic rehabilitation system, surgical video understanding, and climate-health risk forecasting. **Aim to build deployable, reusable AI solutions that create practical business value.**

Core Competencies:

Data Science & Modeling: Python, SQL, PySpark, Statistical Analysis, Feature Engineering, Anomaly Detection

LLM Applications: LangChain, MCP, LLM Agents, Tool-calling Workflows, Structured Output Parsing, Basic RAG

Deep Learning & Computer Vision: PyTorch, TensorFlow, CNNs, Object Detection, Image / Video Understanding, Pose Estimation, Medical Image Processing, Multi-Task Learning, Classification / Ranking Models.

Product & Business Skills: Requirement Decomposition, Business Analysis Workflow Productization, Cross-functional Communication, AI Solution Design, Data-driven Decision Support

EDUCATION

National University of Singapore (Msc.) Biomedical Engineering 2025.08 - 2026.06
Module: Advanced Machine Learning, Robotic Intelligence, Computational Methods, Medical Imaging Systems.

Beijing Institute of Technology (B.Eng.) Biomedical Engineering 2021.08 - 2025.06
Module: Random/Digital Signal Analysis, Probability&Statistics, Machine Learning&AI, Digital Circuits Systems

WORK EXPERIENCE

Tencent Singapore	Tech / Data & AI Intern (IEG Global)	2025.11 - 2026.06
- Built and deployed multiple domain-specific LLM / AI Agents for core game operations scenarios, covering anomaly discovery, attribution analysis, trend interpretation, and conclusion generation, upgrading operational analytics from manual dashboard inspection to natural-language-driven intelligent workflows. - Participated in the design of a modular Agent framework, focusing on Memory management, Skills-based capability abstraction, and tool-calling workflow optimization to improve reusability and scalability across different business analysis scenarios. - Developed an anomaly detection & automated alerting system for core game operation metrics, reducing reliance on manual dashboard monitoring and accelerating abnormality discovery and response. - Built analysis pipelines using SQL, PySpark, BigQuery, and StarRocks database to support large-scale data process		
Didi Global Inc.	Tech / Algorithm Intern (Marketplace Technology)	2025.05 - 2025.08
- Worked on first-order passenger conversion optimization for Didi Hitch by decomposing the transaction funnel into multiple stages, including invitation exposure, passenger acceptance, and final trip completion. - Built Multi-Task Learning models in TensorFlow and designed an ESMM-based framework to jointly model CTV, CVR, and CTCVR, incorporating Bayesian uncertainty weighting and Focal Loss to address sample sparsity, task imbalance, and dependency across conversion stages. - Supported conversion rate optimization for matching, ranking, and pricing strategies by improving the model's ability to capture user intent and multi-stage behavioral signals. - Processed large-scale user behavior, order, and transaction data using Hadoop and PySpark, engineering sequential, categorical, and behavioral intent features for conversion modeling.		

PROJECT EXPERIENCE

Robot Therapist for Assisted Therapeutic Exercises	National University of Singapore	2024.08- 2025.04
- Developed a Robot-assisted rehabilitation system for upper-limb stroke therapy, integrating hybrid robotic arms, an end effector, and MediaPipe / BlazePose-based pose estimation for motion sensing, trajectory tracking, and therapeutic motion replay. - Enhanced motion prediction accuracy via signal filtering + model-based correction, and analyzed sequential tracking data to assess patient compliance and engagement.		
Develop LLM applications based on prompt engineering	NVIDIA Deep Learning Institute	2025.05
- Applied iterative prompt engineering best practices to design and optimize LLM-based applications for language tasks. - Developed end-to-end workflows with LangChain, coding applications for document analysis and generative tasks.		
YOLO-MedTrack	Medical Robotic Intelligence Project @ NUS	2025.11
Real-time Surgical Instrument Segmentation and Tracking System		
- Designed a Python-based computer vision pipeline using YOLOv11-Seg + ByteTrack / MedTrack for robotic-assisted surgical video understanding, enabling surgical instrument instance segmentation, class recognition, cross-frame tracking, and trajectory analysis. - Formulated the task as Video Instance Segmentation, integrating object detection, semantic segmentation, and tracking-by-detection methods to improve temporal consistency under occlusion, scale variation, and low-contrast surgical scenes. - Conducted data preprocessing, model training, and experimental evaluation on the EndoVis 2018 dataset, using mAP, Dice, MOTA, ID Switches, FPS, and latency to analyze the accuracy-speed trade-offs of different detector-tracker combinations.		
Climate-health Risk Forecasting Based on Baidu Index	National Outstanding Project	2023.12 - 2024.03
- Developed an LSTM-ARIMA hybrid model ($R^2=0.922$) for extreme heat prediction across time horizons. - Enhanced short-term forecasting with incremental learning and quantified pollution-health lag effects via DLNM. - Integrated GAM to assess heat-pollution health synergies. Built a multi-city visualization system revealing regional response patterns.		
iGEM 2023 - International Genetically Engineered Machine	Global Golden Medal	2022.12 - 2023.11
Led the Human Practice group on the Oxidative Stress Response Damage Detection System. Directed strategy planning, coordinated with potential users and domain experts, and established a feedback loop to refine project design and improve alignment with real-world needs.		

ADDITIONAL SUMMARY

- Leadership & Communication:** Student Union, college association leadership team, Oxford University study exchange program, and public service practice programs.
- Honors:** Outstanding Undergraduate Student
- MBTI – ENTJ** (The Protagonist): Energetic, Fast learner, empathetic, collaborative, motivating...
- Languages:** English(Proficient), Chinese(Native)
- Hobby:** Sports/ Meditation / Explore / Networking